Fiscal Year:	FY 2015	**Task Last Updated:**	FY 07/15/2015
PI Name:	Turner, Russell T Ph.D.		
Project Title:	Role of Marrow Adipocytes in Bone Loss during Simulated Spaceflight		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Animal Biology: Vertebrate		
Space Biology Cross-Element Discipline:	(1) Musculoskeletal Biology		
Space Biology Special Category:	(1) Translational (Countermeasure) Potential		
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Zip Code:	97331-8558	**Congressional District:**	4
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2011 Space Biology NNH11ZTT002N
Start Date:	09/15/2012	**End Date:**	09/14/2016
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA ARC
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Flight Program:			
Flight Assignment:	NOTE: Extended to 9/14/2016 per A. Chu/ARC and NSSC information (Ed., 7/16/15)		
Key Personnel Changes/Previous PI:	There were no changes in key personnel.		
COI Name (Institution):	Iwaniec, Urszula (Oregon State University) Wong, Carmen (Oregon State University)		
Grant/Contract No.:	NNX12AL24G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Spaceflight, by impairing the differentiation program of stem cells residing within bone marrow results in bone loss, increased bone marrow adiposity, anemia, and impaired immune function. Because these disturbances may potentially compromise the success of long-duration flights, there exists an urgent need to identify the underlying mechanisms and implement effective countermeasures. We propose to test the novel hypothesis that negative energy balance during spaceflight results in 1) bone marrow dysfunction and increased marrow adiposity and 2) altered signaling by peripheral adipokines (cytokines produced by adipocytes) all of which, in turn, contribute to negative bone turnover balance and bone loss. The proposed research will focus on the precise role of adipose tissue and the important adipokine leptin in mediating the detrimental effects of simulated spaceflight (hindlimb unloading) on bone marrow cells in mice by accomplishing the following 2 specific aims: Specific Aim 1 will test the hypothesis that increased bone marrow adiposity plays a causal role in mediating simulated spaceflight-induced changes in bone metabolism. This will be accomplished by comparing the skeletal response to hindlimb unloading of wild type (WT) mice with mice incapable of generating bone marrow adipocytes (kit-receptor-deficient KitW/W-v mice). Specific Aim 2 will test the hypothesis that the adipokine leptin, by targeting the differentiation of bone marrow stem cells, plays a key role in mediating energy sensitive changes in bone metabolism during simulated spaceflight. This will be accomplished by examining whether transplantation of mesenchymal or hematopoietic stem cells from leptin receptor-deficient db/db mice into WT mice alters bone loss during simulated spaceflight. Completion of the proposed project is expected to increase our understanding of the role of energy metabolism in mediating bone loss during simulated spaceflight and suggest specific non-pharmacological and pharmacological interventions to mitigate the detrimental effects of spaceflight on bone marrow.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Osteoporosis is a major health concern. To our knowledge, all forms of osteoporosis involve infiltration of fat into bone marrow. If the increased marrow fat alters the magnitude of bone loss, as hypothesized, this research will have significance to human health that transcends protecting astronauts from bone loss during spaceflight. Our findings suggest that hematopoietic cells inhibit transition of lining cells to osteoblasts. As such, loss of this negative regulation by hematopoietic lineage cells is a plausible mechanism for increased bone turnover following irradiation. An important implication of the putative negative regulation of bone turnover by hematopoietic lineage cells would be that restoration of normal hematopoiesis following irradiation may be essential to restore normal bone turnover. Additionally, blocking the inhibitory actions of hematopoietic cells on osteoblast differentiation may be an attractive approach toward normalizing bone formation during spaceflight. In support of these hypotheses, our findings to date suggest that delivery of a very small number of highly purified hematopoietic stem cells to mice can restore normal hematopoiesis following lethal radiation and limit bone damage. Prevention of morbid obesity in ob/ob mice further exacerbated many of the abnormalities associated with leptin deficiency. Because morbid obesity normally present in ob/ob mice attenuates the abnormal skeletal phenotype associated with leptin deficiency, the role of leptin in regulating postnatal bone growth, maturation, and turnover may have been underestimated. These findings suggest that a reduction in leptin levels due to a negative energy balance during spaceflight could contribute to bone loss in astronauts. Results to date strongly support this hypothesis.
Task Progress:	Progress on Specific Aim 1 Experiment 1 was designed to evaluate differences in skeletal response of wild WT and adipocyte-deficient kitw/w-v mice to simulated spaceflight. The hindlimb unloading portion of the study was begun in Year 1 and completed in Year 2. Progress in Year 2 included: 1) analysis of body composition using duel energy x-ray absorptiometry (DXA); 2) analysis of femur, tibia, and lumbar vertebra microarchitecture using microcomputed tomography (microCT); 3) measurement of serum glucose at necropsy using a glucometer; and 4) embedding femurs in plastic for sectioning and quantitative histomorphometry. Progress in Year 3 included sectioning plastic embedded tissue and performing histological analysis on stained and unstained sections. The results are detailed in Abstracts submitted to American Society of Bone and Mineral Research (ASBMR) annual meeting to be held in Seattle, WA (Oct, 2015) and American Society for Gravitational and Space Research (ASGSR) annual meeting to be held in Alexandria, VA (Nov, 2015). Experiment 2 was designed to evaluate differences in skeletal response of WT mice to simulated spaceflight following transplantation of bone marrow from WT or kitw/w-v mice. As was pointed out in our Year 1 annual report, the radiosensitivity of mice varies markedly among mouse strains. Before performing Experiment 2, we initiated a pilot study in Year 2 to verify feasibility of transplanting hematopoietic stem cells into WT mice. Experiment 2 was modified based on the results of our feasibility study. As noted above, adoptive transfer of Kitw-w/v hematopoietic stem cells into WT mice did not rescue the mice from lethal irradiation. We therefore transferred purified stem cells from WT mice into WT and Kitw-w/v and performed hindlimb unloading following successful bone marrow reconstitution. We performed proposed Experiment 2 in Year 3 and have completed analysis of the bones by DXA and microCT. Progress on Specific Aim 2 Experiment 1 was designed to determine whether a high level of fat within bone marrow is associated with accelerated bone loss during simulated spaceflight. The animal portion of this study was initiated and completed during Year 2. Specifically, weanling (1-month-old) WT and leptin-deficient ob/ob mice were maintained for 3 months under thermoneutral (32ºC) housing conditions to equalize resting energy expenditure between WT and ob/ob mice. The ob/ob mice were pair-fed to the WT mice to prevent development of morbid obesity in the ob/ob mice. Additionally: 1) DXA measurements were performed to measure body composition and the results analyzed; 2) femur, tibia, and lumbar vertebra were scanned ex vivo by microCT and the 3D reconstructions analyzed for determination of bone microarchitecture; 3) serum glucose, P1NP (bone formation marker), osteocalcin (bone turnover marker), and corticosterone (stress hormone) levels were measured at necropsy; and 4) femurs were embedded in plastic for sectioning and quantitative histomorphometry. Progress in Year 3 included sectioning of plastic embedded tissue in preparation for histomorphometric analysis. Experiment 2: This study was designed to test the hypothesis that leptin signaling, acting on mesenchymal stem cells, reduces hindlimb unloading-induced bone loss, whereas leptin signaling, acting on hematopoietic stem cells, promotes bone loss. We have revised this experiment as originally designed based on: 1) our findings in Years 2 and 3 that morbid obesity attenuates the skeletal abnormalities associated with leptin deficiency in mice (J Endocrinol. 2014

	Oct;223(1):M1-15); and 2) new evidence that the effects of leptin to increase bone formation, suppress bone marrow adiposity, and regulate appetite and energy expenditure can be readily disassociated from one another based on circulating levels of the hormone. We will now evaluate the important question as to whether hindlimb unloading blocks the anabolic and antiadipogenic actions of leptin. However, the experiment will also position us to answer our original question. This work will be completed during the no-cost extension. Summary To date we have made excellent progress in testing our hypothesis that negative energy balance during spaceflight results in bone marrow dysfunction and increased marrow adiposity and altered signaling by peripheral adipokines (cytokines produced by adipocytes). These changes contribute to negative bone turnover balance and bone loss. We have completed 3 of the 4 proposed studies (and several additional feasibility studies). We anticipate completing the final hindlimb unloading study (Specific Aim 2) during the no cost extension. Finally, we anticipate that NNX12AL24G will support several additional peer-reviewed publications.
Bibliography Type:	Description: (Last Updated: 01/12/2026)
Articles in Peer-reviewed Journals	Turner RT, Philbrick KA, Wong CP, Olson DA, Branscum AJ, Iwaniec UT. "Morbid obesity attenuates the skeletal abnormalities associated with leptin deficiency in mice." J Endocrinol. 2014 Oct;223(1):M1-15. Epub 2014 Jul 2. http://dx.doi.org/10.1530/JOE-14-0224 ; PubMed PMID: 24990938; PubMed Central PMCID: PMC4161659 , Oct-2014
Articles in Peer-reviewed Journals	Philbrick KA, Turner RT, Branscum AJ, Wong CP, Iwaniec UT. "Paradoxical effects of partial leptin deficiency on bone architecture in growing female mice." Anatomical Record, in press, as of July 2015. , Jul-2015